

Paper Practice 1.6**Descriptive Modeling and Accuracy**

Show all of your work in the space provided. When you use a metric, name it. Explanations should be one clear sentence. Circle your final answers.

- 1.** The school store's hoodie sales for five months were 42, 55, 38, 61, and 49. (Descriptive Modeling)

a. Find the mean number of hoodies sold per month.

b. Name one thing the mean HIDES about these sales.

c. Would a table or a bar graph be more useful for spotting a trend? Explain.

- 2.** Two players' season statistics are shown.
(Compare Metrics)

Player	Points	Games
Alvarez	1,848	22
Brooks	1,530	17

a. Who leads in total points?

b. Who leads in points per game?

c. Which metric would you use to argue for each player? What does this show?

- 3.** A car travels 260 miles on 8 gallons of gas. Find the miles per gallon, and explain what that metric tells a buyer. (Metrics)

- 4.** A digital scale reads 9.985 grams for a mass whose true value is 10 grams. Find the accuracy of the reading. (Accuracy)

- 5.** A bag of flour labeled 5 pounds weighs 4.92 pounds. Find the accuracy, then express it as a percent of the labeled weight. (Accuracy)

- 6.** Explain why accuracy is written with absolute value bars rather than plain subtraction.
(Explain)

- 7.** A principal wants to name the “most improved” class. Class A rose from 62 to 74; Class B rose from 80 to 88. (Choose a Metric)

a. Find each class's raw point gain. Who improved more?

b. Find each class's percent gain. Who improved more?

c. Both metrics agree here. Why does that make the conclusion stronger?

- 8.** Error Analysis – Gus says a thermometer reading of 101.2°F when the true temperature is 101.0°F has an accuracy of -0.2°F . Identify his mistake. (Error Analysis)

- 9.** A team scored 1848 points in 22 games. Find the points per game, then explain why a coach might prefer this metric over total points. (Metrics)

- 10.** Describe a real situation where the mean would be a POOR model, and name a better metric. (Reason)